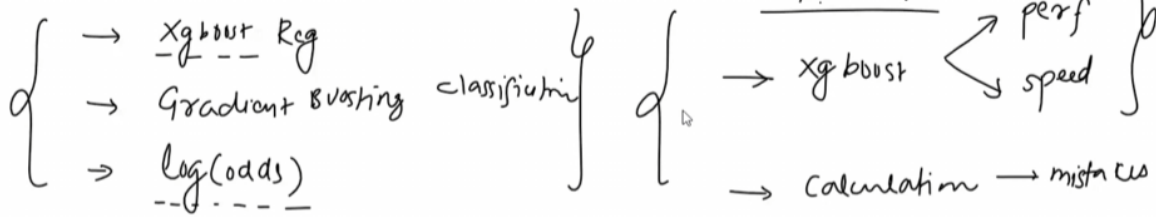


Prerequisite & Disclaimer

11 November 2023 09:29



Step by Step Solution

11 November 2023 17:30

cgpa	placed?	pred 1(10)
5.70	0	0.405 ✓
6.25	1	0.405
7.10	0	0.405
8.15	1	0.405
9.60	1	0.405

Stage 1 - $\square \rightarrow \log(\text{odds}) \rightarrow \text{prob}$

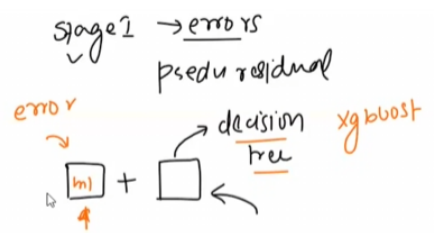
base estimator $\rightarrow$ mean

$\log(\text{odds}) \rightarrow \log\left(\frac{p}{1-p}\right)$ $p \rightarrow \text{prob +ve class (1)}$

$$\log\left(\frac{3/5}{2/5}\right) = \log_e\left(\frac{3}{2}\right) = 0.405$$

$$p = \frac{e^{\log \text{odds}}}{1 + e^{\log \text{odds}}} = e^{0.405}$$

cgpa	placed?	pred 1 (lo)	pred 1 (prob)	res1
5.70	0	0.405 ✓	0.6	-0.6 ← predict
6.25	1	0.405	0.6	0.4
7.10	0	0.405	0.6	-0.6
8.15	1	0.405	0.6	0.4
9.60	1	0.405	0.6	0.4



	<u>cgpav</u>	res1
① 5.97	5.70	-0.6 ←
② 6.67	6.25	0.4
③ 7.62	7.10	-0.6
④ 8.87	8.15	0.4
⑤ 8.87	9.60	0.4

leaf node

→ $-0.6, 0.4, -0.6, 0.4, 0.4$ → similarity score

→ $SS = 0$

maximize

GBDT classifier

$$SS = (\sum \text{residual}_i)^2 \rightarrow$$

$$\sum \text{prev-prob}_i (1 - \text{prev-prob}_i) + \frac{\lambda}{2} \rightarrow \text{reg } \lambda = 0$$

$$= \frac{(-0.6 + 0.4 - 0.6 + 0.4 + 0.4)^2}{5 \times [0.6(1-0.6)] + 0} = 0$$

cgpa	res1
5.70	-0.6
6.25	0.4
7.10	-0.6
8.15	0.4
9.60	0.4

Splitting criteria = 5.97

$cgpa < 5.97$ ✓

Left Branch ($cgpa < 5.97$)	Right Branch ($cgpa \geq 5.97$)
-0.6	0.4, -0.6, 0.4, 0.4
SS_L	SS_R
$\frac{(-0.6)^2}{0.6(1-0.6)}$	$\frac{(0.4 - 0.6 + 0.4 + 0.4)^2}{4 \times 0.6 \times 0.4}$
$= \frac{0.36}{0.24}$	$= \frac{0.36}{4 \times 0.24} = \frac{1.5}{4}$
$= 1.5$	$= 0.37$

gain = $(SS_L + SS_R) - SS_{root}$
 $= (1.5 + 0.37) - 0$
 $= 1.87$

Splitting criteria = 6.67

$cgpa < 6.67$ ✓

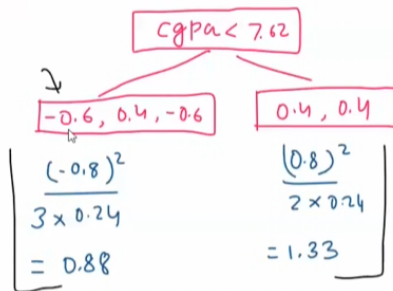
Left Branch ($cgpa < 6.67$)	Right Branch ($cgpa \geq 6.67$)
-0.6, 0.4	-0.6, 0.4, 0.4
SS_L	SS_R
$\frac{(-0.2)^2}{2[0.6 \times 0.4]}$	$\frac{(0.2)^2}{3 \times 0.24}$
$= \frac{0.04}{2 \times 0.24}$	$= \frac{0.04}{3 \times 0.24}$
$= \frac{0.04}{0.48} = 0.08$	$= 0.05$

gain = $0.08 + 0.05 - 0$
 $= 0.13$



<u>cgpa</u>	res1
5.97	5.70
6.67	6.25
7.62	7.10
8.87	8.15
9.60	9.60

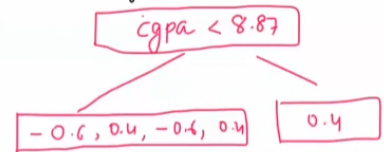
splitting criteria = 7.62



$$\text{gain} = 0.88 + 1.33 - 0$$

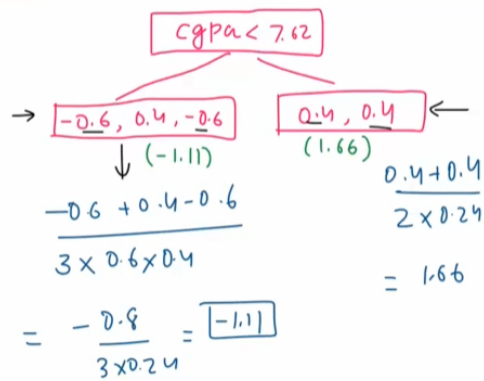
$$= \boxed{2.22} \rightarrow$$

splitting criteria = 8.87



stage 2 model

output

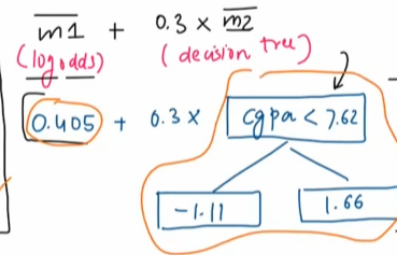


$$\text{Output} = \frac{\sum \text{residual}_i}{\sum \text{prev-prob}_i (1 - \text{prev-prob}_i) + \lambda} = 0$$

stage 11 → prediction

	cgpa	placed?	pred1(lo)	pred1(prob)	res1	pred2(lo)	pred2(prob)	res2
→	5.70	0	0.405	0.6	-0.6	0.072	0.518	-0.51
→	6.25	1	0.405	0.6	0.4	0.072	0.518	0.48
→	7.10	0	0.405	0.6	-0.6	0.072	0.518	-0.51
→	8.15	1	0.405	0.6	0.4	0.903	0.712	0.28
→	9.60	1	0.405	0.6	0.4	0.903	0.712	0.28

Stage 2 model



$$0.405 + 0.3 \times (-1.11)$$

etc

$$0.405 + 0.3 \times (-1.11)$$

$$0.405 + 0.3 \times 1.66$$

$m3 \rightarrow$ decision

log(odds)
→ prob

$$\frac{e^{\log odds}}{1 + e^{\log odds}}$$

Stage 3

$[m1 + 0.3 \times m2 + 0.3 \times m3] \rightarrow \text{pred} \rightarrow \text{prob} \rightarrow \text{res3}$
 cgpa/res3 → my decision